

Final Project Report Template

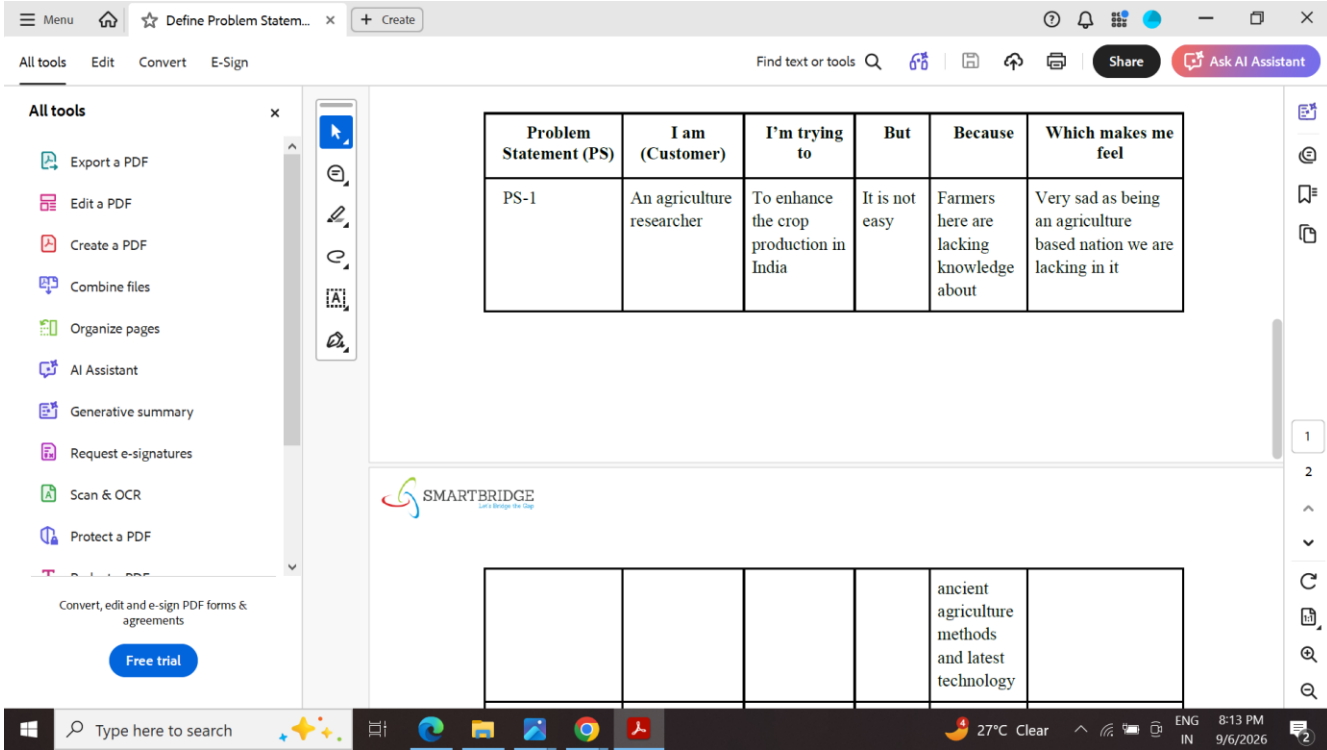
1. Introduction

1.1. India's Agricultural Production Analysis

1.2. To analyze the different crops in different states and seasons of India.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement



The screenshot shows the SMARTBRIDGE web application interface. The top navigation bar includes a 'Menu' button, a 'Define Problem Statement...' tab, and a '+ Create' button. Below the navigation bar, there are tabs for 'All tools', 'Edit', 'Convert', and 'E-Sign'. A search bar labeled 'Find text or tools' is present, along with a 'Share' button and an 'Ask AI Assistant' button.

On the left side, there is a sidebar titled 'All tools' with a list of tools: Export a PDF, Edit a PDF, Create a PDF, Combine files, Organize pages, AI Assistant, Generative summary, Request e-signatures, Scan & OCR, and Protect a PDF. Below this list is a 'Free trial' button.

The main content area displays a table with the following structure:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An agriculture researcher	To enhance the crop production in India	It is not easy	Farmers here are lacking knowledge about	Very sad as being an agriculture based nation we are lacking in it

Below the table, there is a SMARTBRIDGE logo and a smaller table with the following structure:

				ancient agriculture methods and latest technology	
--	--	--	--	---	--

The bottom of the screenshot shows a Windows taskbar with a search bar, several application icons, and a system tray displaying the date and time (8:13 PM, 9/6/2026).

2.2. Project Proposal (Proposed Solution)

Menu

Define Problem Statements Te...

Project Proposal (Prop...

Create

All tools

Edit

Convert

E-Sign

Find text or tools

Share

Ask AI Assistant

Export a PDF

Edit a PDF

Create a PDF

Combine files

Organize pages

AI Assistant

Generative summary

Request e-signatures

Scan & OCR

Generative summary

Request e-signatures

Scan & OCR

Protect a PDF

Convert, edit and e-sign PDF forms & agreements

AI Assistant

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Free trial

Project Overview

Objective

To study and analyze crop production in different states of India and understand the changes in crop production over the years

Scope

The project covers crop production data from different states and years. It will show crop-wise, state-wise production using charts and dashboards

Problem Statement

Description

India produces many types of crops in different states. It is difficult to understand and compare all the crop production data manually. A simple data anlysis and visualization system is needed to understand the data easily.

Impact

This analysis can help users understand which crops are produced more, which states have higher production, and how crop production changes over time.

Proposed Solution

Approach

First, collect the agriculture production data. Then clean and organize the data. After that, analyze the data and crests chats and an interactive dashboard to show the results clearly.

Key Features

1. State-wise crop production

2. Crop-wise production

Software

Frameworks

Tools used for analysis and visualization

Tableau

Libraries

Libraries for data cleaning/analysis

Python, Pandas

Development Environment

Software used for project work

Tableau/ Jupyter Notebook

Data

Data

Agricultural crop production dataset

CSV/ Excel dataset

1 of 2

2 of 2

2 of 2

2.3. Initial Project Planning

Menu

Define Problem Statements Te...

Project Proposal (Proposed Sol...

Project Planning Templ...

Create

All tools

Edit

Convert

E-Sign

Find text or tools

Share

Ask AI Assistant

Save time with a document summary using AI Assistant.

View Summary

SMARTBRIDGE

Initial Project Planning Template

Date	06 September 2026
Team ID	
Project Name	India's Agricultural Production Analysis
Maximum Marks	4 Marks

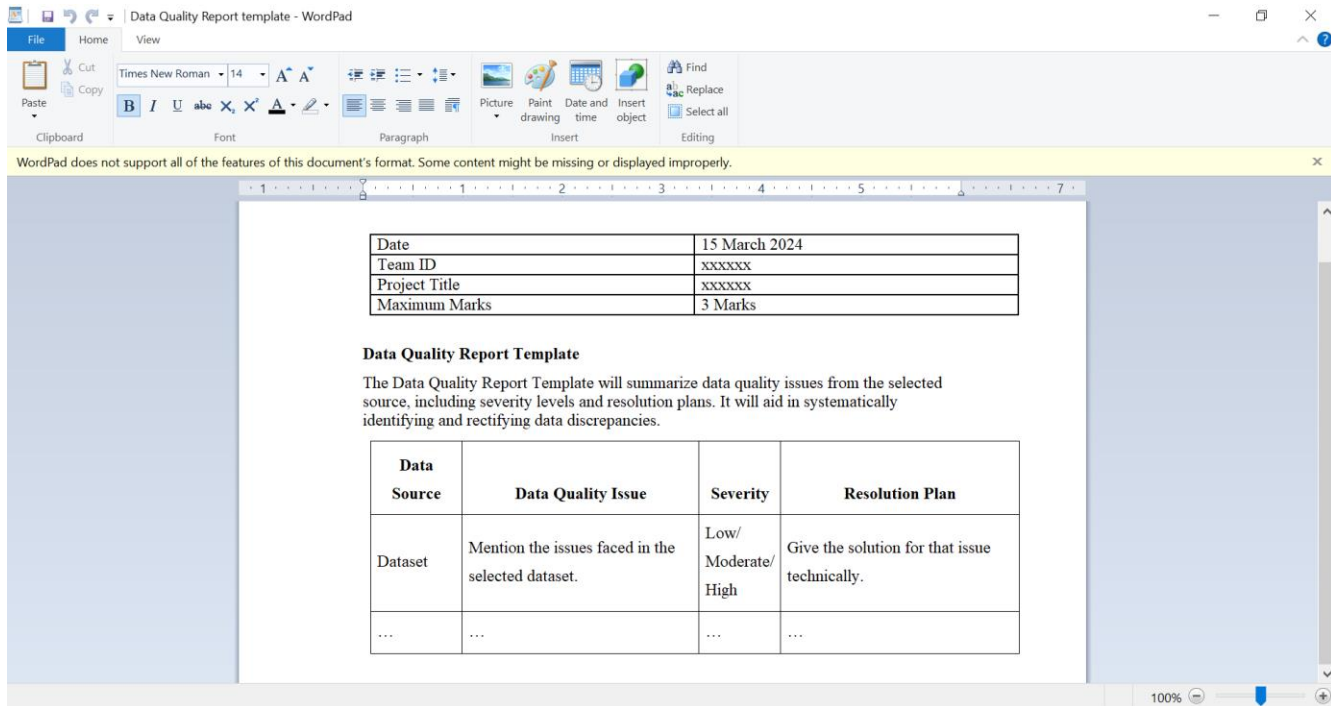
Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection and extraction	USN-1	Story 1. Downloading dataset Story 2. Loading the dataset	2 1	Easy Very easy	Krishn Kumar Singh	20 August 2026	29 August 2026
Sprint-1	Data Preparation	USN-2	Story 3. Clean the data Story 6. Creating story	3 3	Moderate Moderate	Krishn Kumar Singh	20 August 2026	29 August 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified



3.3. Data Exploration and Preprocessing

Define Problem State...Project Proposal (Propos...Project Planning Templa...Raw Data Sources Ident...Data Exploration and P...Create

All toolsEditConvertE-SignFind text or toolsShareAsk AI Assistant

All tools

Export a PDF

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Combine files

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Free trial

Get an instant AI-powered summary whenever you open a docume...Generative AI User GuidelinesTurn on auto-summary

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Data Collection and Preprocessing Phase

Date	06 September 2026
Team ID	
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The data consists of 345,408 rows and 10 columns that correspond to different agriculture metrics spanning the years 1997-2021. The fields are State, Crop, District, Year, Season which are dimensions and Area, Area units, Production, Production units and Yields are measures.
Data Cleaning	Missing values: Isolated and deleted incomplete rows to prevent statistical skews.
Data Transformation	Calculated Fields: We created a new calculated field "Yield" which is the ratio of production per unit area. We also used filters for specifying a visual for a particular state and many other such filters.
Data Type Conversion	Validated that state, crop, district year, and season are classified as non-numeric strings. Numerical values like area, production are in million formats.
Column Splitting and Merging	Not needed.
Data Modeling	Not needed.

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2 of 2

2 of 2

4. Data Visualization

4.1. Framing Business Questions

4.2. Developing Visualizations

5. Dashboard

Define Problem St...

Project Proposal

Project Planning T...

Raw Data Sources

Data Exploration a...

Business Question and ...

Create

All tools

Convert

E-Sign

Find tool or tools

Share

Add AI Assistant

Get an instant AI-powered summary whenever you open a document... Generative AI User Guidelines

Turn on auto-summary

SMARTBRIDGE

Business Question and Visualization Report

Date	06 September 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

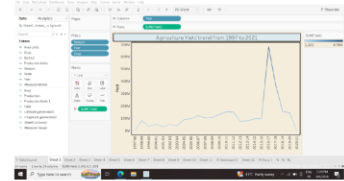
Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

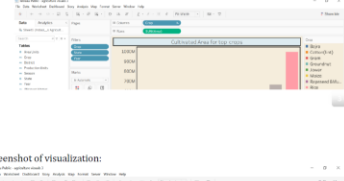
1. What are agriculture yield trends over the years?

- Visualization: Line chart showing yearly agriculture yield trends.
- Screenshot of visualisation



3. Which crops are most produced across India?

- Visualization: Bar chart showing top crops by area.
- Screenshot of visualisation



- Screenshot of visualization:



5.1. Dashboard Design File

Dashboard Design

Date	06 September 2026
Team ID	
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	5 Marks

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design:

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

Note: Highlight the major outcomes in form of bullet points

Sample:

Here are five potential outcomes from the dashboard image provided:

- **Highest yielding year:** 2016-17 is the year in which highest agriculture yield of about 679M is obtained.
- **Highest Production:** Kerala is the state with highest crop production and Ladakh is the state with lowest crop production.
- **Peak season:** Kharif Season has the highest production while autumn has the lowest crop production.
- **Top yielding crop:** Sugarcane and Sesamum are the most yielding crops in whole India.
- **Cultivable area for crops:** Rice and wheat are the highest cultivated crops in whole of India.

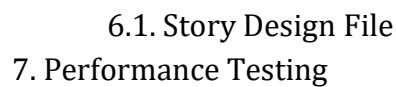
Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

Example

https://public.tableau.com/views/agriculturevisuals3/Dashboard1?language=en-US&publish=yes&sid=&redirect-auth&display_count=n&originviz_share_link

6. Report



7. Performance Testing

- 7.1 Total 5 filters used: state filter, year filter, crop filter, season filter, production range filter. All filters are working correctly in the dashboard. Average production

7.2 one calculation field – $\text{yield} = (\text{production} / \text{area})$

1. 7.3 Total visualizations are 9
2. Total dashboard 1
3. Total story 1

8. Conclusion/Observation

1. Rice is the most cultivated crop in India in terms of area.
2. Karnataka, Tamil Nadu, Madhya Pradesh and West Bengal are leading states in production.
3. Kharif season contributes more than 50% of total production (approx. 720M tones).
4. Total agriculture production has shown an increasing trend from 1997 to 2019.
5. The data for 2020-21 is incomplete and needs to be corrected.
6. The dashboard provides a one-stop solution for agricultural analysis.

9. Future Scope

1. Add weather and rainfall data to predict production.
2. Use forecasting to predict production for the next 5 years.
3. Add district level drill down analysis.
4. Connect the dashboard to a live government API for real-time updates.
5. Create a mobile-friendly version of the dashboard.